INCH-POUND

MIL-T-27/213B
20 April 1993
SUPERSEDING
MIL-T-27/213A
18 September 1980

MILITARY SPECIFICATION SHEET

TRANSFORMER, POWER, STEP-UP, 178 VOLTAMPERES, TF5TO3ZZ

(B) Inactive for new design after the date of this specification

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-T-27

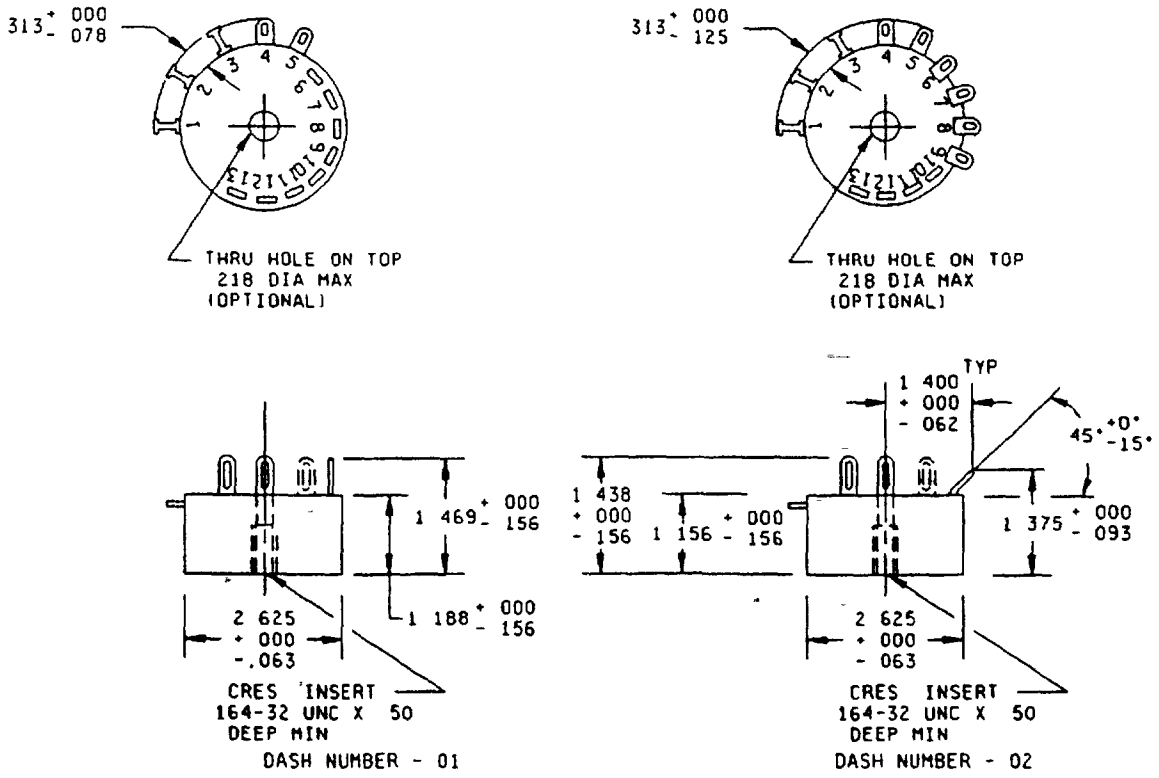


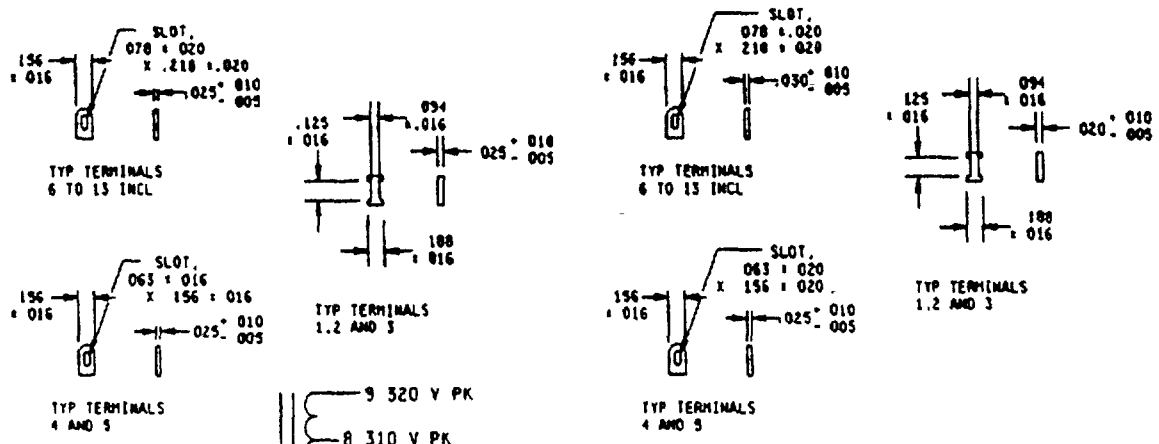
FIGURE 1 Dimensions and configurations

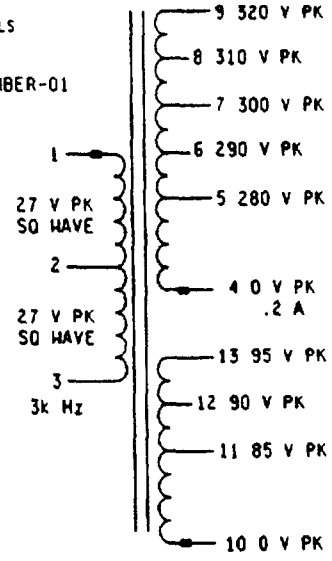
(B) denotes changes

FSC 5950

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WORKING VOLTAGE (PEAK)
(1-2)=27 V
(4-9)=320 V
(10-13)=95 V

CIRCUIT DIAGRAM AND MARKING

Inches	mm	Inches	mm
.005	0 13	164	4 17
010	0.25	188	4.78
016	0.41	218	5 54
020	0 51	.313	7 95
025	0 64	500	12 70
.062	1.57	1 156	29.35
063	1 60	1 188	30 18
078	1 98	1 375	34 92
093	2.36	1 400	35.56
094	2.39	1 438	36 53
125	3.18	1 469	37.31
156	3 96	2 625	66.68

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for information only
3. Terminal spacing 30° ±2° between 1-2, 2-3, and 3-4, 22 5° ±2° between all others located on 2.375 (60 32 mm) ± 032 (0 81 mm) inch diameter circle
4. Marking shall be on the sides of the case

FIGURE 1 Dimensions and configurations - Continued

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REQUIREMENTS (When numbers in parentheses, i e , (1-2) are used, they indicate the winding and the extreme terminals of the windings)

Electrical ratings

Primary voltage (1-3) 54 volts peak ct, 3 kHz square wave

Primary current (1-3) 3 3 amperes peak

Voltampere rating 178 voltamperes

Secondary voltage

(4-5)	280 volts peak
(4-6)	290 volts peak
(4-7)	300 volts peak
(4-8)	310 volts peak
(4-9)	320 volts peak
(10-11)	85 volts peak
(10-12)	90 volts peak
(10-13)	95 volts peak

Secondary current

(4-9)	0 2 ampere rms
(10-13)	1 2 amperes rms

Working voltage (peak)

(1-2)	27 volts
(4-9)	320 volts
(10-13)	95 volts

Design and construction

Dimensions and configuration See figure 1

Duty cycle Continuous

Case Encapsulated

Material Epoxy

Terminals All terminals to be cartridge brass, 70 percent copper and 30 percent zinc in accordance with QQ-B-613

Finish Hot-tinned finish (60 percent tin and 40 percent lead) conforming to QQ-S-571

Height

1-2-3-4-5	313 + 000, - 125 inch
6-7-8-9-10-11-12-13	281 + 000, - 156 inch

Weight 315 grams maximum

Operating temperature range -55°C to +170°C

Terminal strength MIL-STD-202, method 211, test condition A

Terminals 1,2,3	5 pounds
Terminals 4 through 13	2 pounds

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Dielectric withstanding voltage (at sea level) 1,000 volts rms

Electrical characteristics

No load With 54 volts peak ct and 3 kHz square wave across (1-3)

Current in (1-3) 0 130 ampere

Power in (1-3) 7 02 watts

Voltage across

(4-5)	288 volts peak ±3 percent
(4-6)	298 volts peak ±3 percent
(4-7)	300 volts peak ±3 percent
(4-8)	318 volts peak ±3 percent
(4-9)	326 volts peak ±3 percent
(10-11)	87 5 volts peak ±3 percent
(10-12)	92 5 volts peak ±3 percent
(10-13)	98 volts peak ±3 percent

Rated load With 54 volts peak ct and 3 kHz square wave across (1-3)

Current in (1-3) 3 3 amperes peak

Power in (1-3) 178 watts

Voltage across

(4-5)	280 volts peak ±3 percent
(4-6)	290 volts peak ±3 percent
(4-7)	300 volts peak ±3 percent
(4-8)	310 volts peak ±3 percent
(4-9)	320 volts peak ±3 percent
(10-11)	85 volts peak ±3 percent
(10-12)	90 volts peak ±3 percent
(10-13)	95 volts peak ±3 percent

Current in

(4-9)	0 2 ampere
(10-13)	1 2 amperes

DC resistance

(1-3)	0 118 to 0 16 ohm
(4-9).	11 90 to 16 10 ohms
(10-13)	0 51 to 0 69 ohm

Polarity Additive with terminals 3 and 4, and 9 and 10 connected

Temperature rise 70°C with 54 volts peak and 3 kHz square wave across (1-3) at an ambient temperature of 100°C

Shock (specified pulse) MIL-STD-202, method 213, test condition H

(B) Quality assurance provisions

Qualification inspection Not applicable for this specification sheet

Quality conformance inspection Groups A and B tests of MIL-T-27 shall be applicable

Marking location See figure 1

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Part or Identifying Number (PIN) M27/213-(dash number from figure 1)

CONCLUDING MATERIAL

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - AR
Navy - OS, SH
Air Force - 17, 99
DLA - ES

User activities
Army - ME
Navy - AS, MC
Air Force - 19

Preparing activity
Army - ER

Agent
DLA - ES

(Project 5950-0815-22)

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